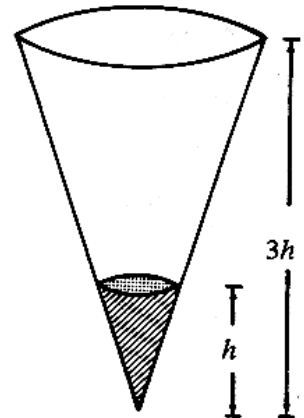


**1994 HKCEE MATHS Paper II**

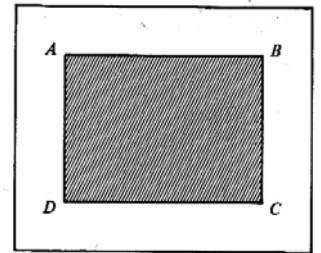
- 1 If  $f(x) = x^2 + 2x$ , then  $f(x-1) =$
- A.  $x^2$   
 B.  $x^2 - 1$   
 C.  $x^2 + 2x - 1$   
 D.  $x^2 + 2x - 3$   
 E.  $x^2 + 4x - 1$
- 2 If  $y = \frac{2x-1}{x+2}$ , then  $x =$
- A.  $\frac{1+3y}{2}$   
 B.  $\frac{1+2y}{2+y}$   
 C.  $\frac{1+2y}{2-y}$   
 D.  $\frac{1-2y}{2+y}$   
 E.  $\frac{1-2y}{2-y}$
- 3 The L.C.M. of  $(x-1)^2$ ,  $x^2 - 1$  and  $x^3 - 1$  is
- A.  $x - 1$   
 B.  $(x-1)^4(x+1)(x^2+x+1)$   
 C.  $(x-1)^2(x+1)(x^2+x+1)$   
 D.  $(x-1)^2(x+1)(x^2-x+1)$   
 E.  $(x-1)(x+1)(x^2+x+1)$
- 4 If  $a = \sqrt{3} + \sqrt{2}$ , then  $a - \frac{1}{a} =$
- A. 0  
 B.  $2\sqrt{2}$   
 C.  $2\sqrt{3}$   
 D.  $\sqrt{3} - \sqrt{2}$   
 E.  $\frac{2\sqrt{3}}{3} + \frac{\sqrt{2}}{2}$
- 5 In the figure,  $(x, y)$  is a point in the shaded region (including the boundary) and  $x, y$  are integers. Find the greatest value of  $3x + y$ .
- A. 7  
 B. 8  
 C. 9.2  
 D. 10  
 E. 10.5
- 
- 6 If  $x(x+1) < 5(x+1)$ , then
- A.  $x < 5$   
 B.  $x < -5$  or  $x > 1$   
 C.  $x < -1$  or  $x > 5$   
 D.  $-5 < x < 1$   
 E.  $-1 < x < 5$
- 7 Which of the following is/are an identity /identities?
- I.  $(x+2)(x-2) = x^2 - 4$   
 II.  $(x+2)(x-2) = 0$   
 III.  $(x+2)^3 = x^3 + 8$
- A. I only  
 B. II only  
 C. III only  
 D. I and III only  
 E. II and III only
- 8 If  $\alpha \neq \beta$  and  $\begin{cases} 3\alpha^2 - h\alpha - b = 0 \\ 3\beta^2 - h\beta - b = 0 \end{cases}$ , then  $\alpha + \beta =$
- A.  $-\frac{b}{3}$   
 B.  $\frac{b}{3}$   
 C.  $h$   
 D.  $-\frac{h}{3}$   
 E.  $\frac{h}{3}$
- 9 Mr. Chan bought a car for \$143 900. If the value of the car goes down by 10% each year, find its value at the end of the third year. (Give your answer correct to the nearest hundred dollars.)
- A. \$94 400  
 B. \$100 700  
 C. \$104 900  
 D. \$115 100

- E. \$116 600
- 10 A wholesaler sells an article to a retailer at a profit of 20%. The retailer sells it to a customer for \$3 600 at a profit of \$720. Find the original cost of the article to the wholesaler.
- A. \$2 304  
B. \$2 400  
C. \$2 880
- D. \$3 000  
E. \$3 456
- 11 The bearing of  $A$  from  $B$  is  $075^\circ$ . What is the bearing of  $B$  from  $A$ ?
- A.  $015^\circ$   
B.  $075^\circ$   
C.  $105^\circ$
- D.  $195^\circ$   
E.  $255^\circ$
- 12 If the sum to infinity of a G.S. is  $\frac{81}{4}$  and its second term is  $-9$ , the common ratio is
- A.  $-\frac{1}{3}$   
B.  $\frac{1}{3}$   
C.  $-\frac{4}{3}$
- D.  $\frac{4}{3}$   
E.  $-\frac{4}{9}$

- 13 In the figure, the paper cup in the form of a circular cone contains 10ml of water. How many ml of water must be added to fill up the paper cup?

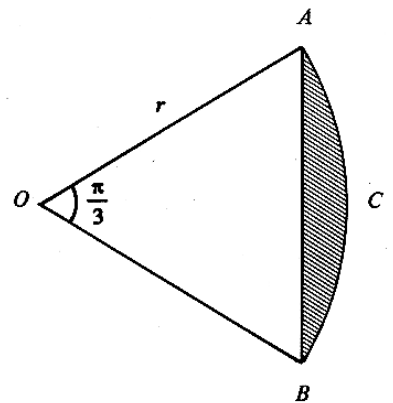


- A. 20  
B. 80  
C. 90  
D. 260  
E. 270
- 14 In the figure,  $ABCD$  is a rectangular field of length  $p$  metres and width  $q$  metres. The path around the field is of width 2 metres. Find the area of the path.



- A.  $(4p + 4q)m^2$   
B.  $(2p + 2q + 4)m^2$   
C.  $(2p + 2q + 16)m^2$   
D.  $(4p + 4q + 16)m^2$   
E.  $(pq + 4p + 4q + 16)m^2$
- 15 In the figure,  $OACB$  is a sector of radius  $r$ .

If  $\angle AOB = \frac{\pi}{3}$ , find the area of the shaded part.



- A.  $\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)r^2$   
B.  $\left(\frac{\pi}{6} - \frac{1}{4}\right)r^2$   
C.  $\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right)r^2$

D.  $\left(\frac{\pi}{3} - \frac{1}{2}\right)r^2$

E.  $\frac{\pi}{3}r - \frac{\sqrt{3}}{4}r^2$

16  $\frac{\cos \theta}{\sin \theta + 1} - \frac{\cos \theta}{\sin \theta - 1} =$

A.  $\frac{2}{\cos \theta}$

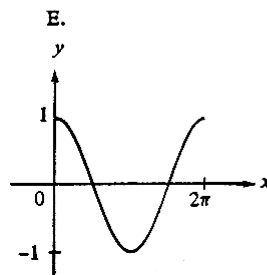
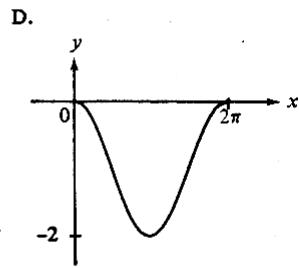
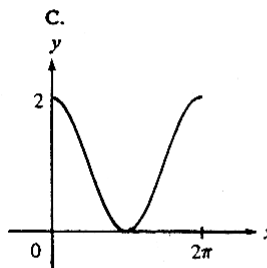
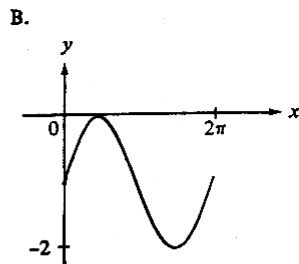
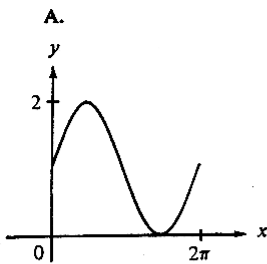
B.  $-\frac{2}{\cos \theta}$

C. 0

D.  $2 \tan \theta$

E.  $-2 \tan \theta$

- 17 Which of the following figures shows the graph of  $y = 1 + \sin x$ ?



18  $\frac{\sin(180^\circ + \theta)}{\cos(90^\circ - \theta)} =$

A.  $\tan \theta$

- 19 In the figure,  $ABCD$  is a cyclic quadrilateral with  $AB=5$ ,  $BC=2$  and  $\angle ADC=120^\circ$ . Find  $AC$ .

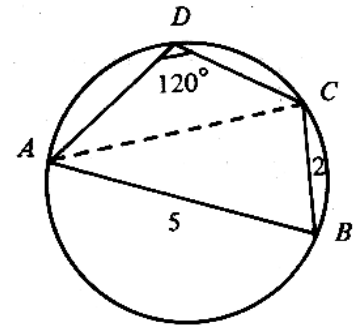
A.  $\sqrt{19}$

B.  $\sqrt{21}$

C.  $2\sqrt{6}$

D.  $\sqrt{34}$

E.  $\sqrt{39}$



- 20 In the figure,  $PC$  is a vertical pole standing on the horizontal plane  $ABC$ . If  $\angle ABC=90^\circ$ ,  $\angle BAC=30^\circ$ ,  $AC=6$  and  $PC=5$ , find  $\tan \theta$ .

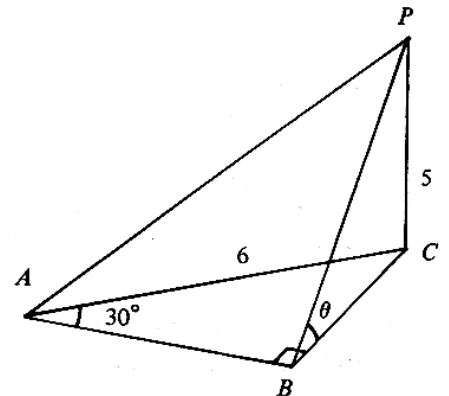
A.  $\frac{3}{5}$

B.  $\frac{5}{6}$

C.  $\frac{5}{3}$

D.  $\frac{3\sqrt{3}}{5}$

E.  $\frac{5\sqrt{3}}{9}$

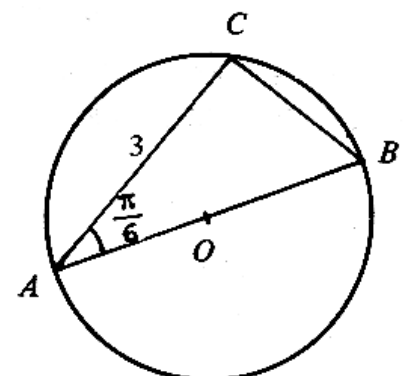


- 21 In the figure,  $O$  is the center of the circle. If  $AC=3$  and  $\angle BAC=\frac{\pi}{6}$ , find the diameter  $AB$ .

A.  $\frac{3}{2}$

B. 6

C.  $\frac{3\sqrt{3}}{2}$



D.  $2\sqrt{3}$

E.  $3\sqrt{3}$

- 22 In the figure,  $PA$  is tangent to the circle at  $A$ ,  $\angle CAP = 28^\circ$  and  $BA = BC$ . Find  $x$ .

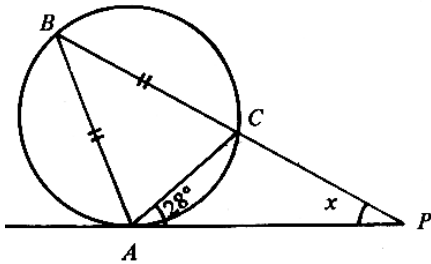
A.  $28^\circ$

B.  $48^\circ$

C.  $56^\circ$

D.  $62^\circ$

E.  $76^\circ$



- 23 In the figure,  $O$  is the center of the inscribed circle of  $\triangle ABC$ . If  $\angle OAC = 30^\circ$  and  $\angle OCA = 25^\circ$ , find  $\angle ABC$ .

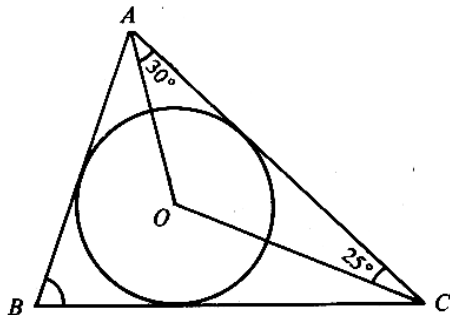
A.  $50^\circ$

B.  $55^\circ$

C.  $60^\circ$

D.  $62.5^\circ$

E.  $70^\circ$



- 24 In the figure,  $AB = AD$  and  $BC = CD$ . If  $\angle BAD = 80^\circ$  and  $\angle ADC = 65^\circ$ , then  $\angle BCD =$

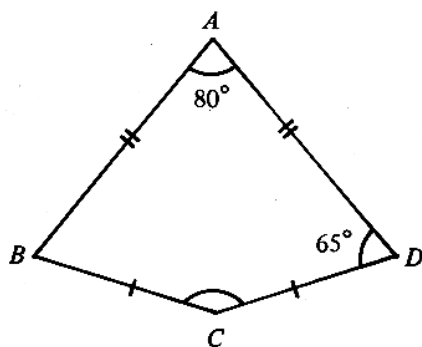
A.  $100^\circ$

B.  $130^\circ$

C.  $145^\circ$

D.  $150^\circ$

E.  $160^\circ$



- 25 In the figure,  $x$ ,  $y$  and  $z$  are the exterior angles of  $\triangle ABC$ . If  $x : y : z = 4 : 5 : 6$ , then  $\angle BAC =$

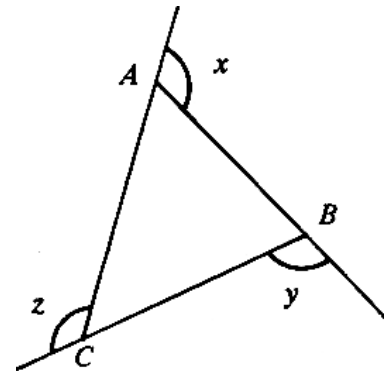
A.  $48^\circ$

B.  $84^\circ$

C.  $96^\circ$

D.  $120^\circ$

E.  $132^\circ$



- 26 The points  $A(4, -1)$ ,  $B(-2, 3)$  and  $C(x, 5)$  lie on a straight line. Find  $x$ .

A.  $-5$

B.  $-4$

D.  $2$

C.  $0$

E.  $3$

- 27 In the figure, the shaded part is bounded by the axes, the lines  $x = 3$  and  $x + y = 5$ . Find its area.

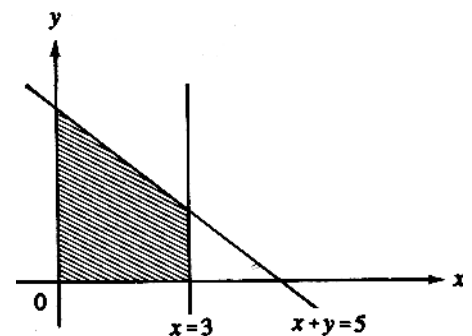
A.  $10.5$

B.  $12$

C.  $15$

D.  $19.5$

E.  $21$



- 28  $AB$  is a diameter of the circle  $x^2 + y^2 - 2x - 2y - 18 = 0$ . If  $A$  is  $(3, 5)$ , then  $B$  is

A.  $(2, 3)$

- B.  $(1, -1)$                       D.  $(-5, -7)$   
 C.  $(-1, -3)$                       E.  $(-7, -9)$

29 The equations of two circles

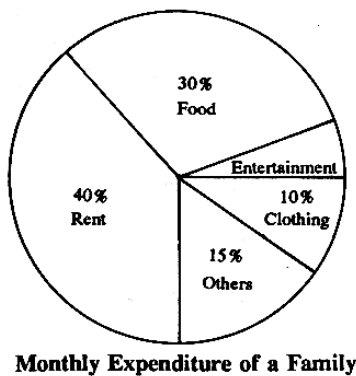
$$\text{are } \begin{cases} x^2 + y^2 - 4x - 6y = 0 \\ x^2 + y^2 + 4x + 6y = 0 \end{cases}$$

Which of the following is/are true ?

- I. The two circles have the same center.  
 II. The two circles have equal radii.  
 III. The two circles pass through the origin.

- A. I only  
 B. II only                      D. I and III only  
 C. III only                      E. II and III only

30 In the figure, the pie chart shows the monthly expenditure of a family. If the family spends \$4800 monthly on rent, what is the monthly expenditure on entertainment ?



- A. \$240  
 B. \$600  
 C. \$720  
 D. \$1 800  
 E. \$12 000

31 A box contains 5 eggs, 2 of which are rotten. If 2 eggs are chosen at random, find the probability that exactly one of them is rotten.

- A.  $\frac{2}{5}$   
 B.  $\frac{3}{5}$                       D.  $\frac{6}{25}$

32 The mean, standard deviation and interquartile range of  $n$  numbers are  $m$ ,  $s$  and  $q$  respectively. If 3 is added to each of the  $n$  numbers, what will be their new mean, standard deviation and interquartile range ?

|    | Mean    | Standard Deviation | Interquartile Range |
|----|---------|--------------------|---------------------|
| A. | $m$     | $s$                | $q$                 |
| B. | $m$     | $s + 3$            | $q + 3$             |
| C. | $m + 3$ | $s$                | $q$                 |
| D. | $m + 3$ | $s$                | $q + 3$             |
| E. | $m + 3$ | $s + 3$            | $q + 3$             |

33  $(3^x)^2 =$

- A.  $3^{(x^2)}$   
 B.  $3^{x+2}$                       D.  $6^x$   
 C.  $3^{2x}$                       E.  $9^{2x}$

34 If  $\log 2 = a$  and  $\log 9 = b$ , then  $\log 12 =$

- A.  $2a + \frac{b}{3}$   
 B.  $2a + \frac{b}{2}$                       D.  $a^2 + b^{\frac{1}{2}}$   
 C.  $\frac{2}{3}a + \frac{2}{3}b$                       E.  $a^2 b^{\frac{1}{2}}$

35 Factorize  $a^2 - 2ab + b^2 - a + b$ .

- A.  $(a-b)(a-b-1)$   
 B.  $(a-b)(a-b+1)$

- C.  $(a-b)(a+b-1)$   
 D.  $(a+b)(a-b+1)$   
 E.  $(a-b-1)^2$

36 
$$\frac{\frac{2}{x} - \frac{1}{y}}{\frac{4y}{x} - \frac{x}{y}} =$$

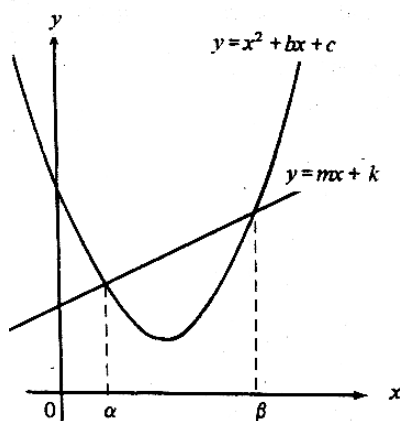
- A.  $2y-x$   
 B.  $2y+x$   
 C.  $\frac{1}{2y-x}$   
 D.  $\frac{1}{2y+x}$   
 E.  $\frac{1}{4y-x}$

- 37  $P(x)$  is a polynomial. When  $P(x)$  is divided by  $(5x-2)$ , the remainder is  $R$ . If  $P(x)$  is divided by  $(2-5x)$ , then the remainder is

- A.  $R$   
 B.  $-R$   
 C.  $\frac{2}{5}R$   
 D.  $\frac{2}{5}$   
 E.  $-\frac{2}{5}$

- 38 In the figure, the line  $y = mx + k$  cuts the curve  $y = x^2 + bx + c$  at  $x = \alpha$  and  $x = \beta$ . Find the value of  $\alpha\beta$ .

- A.  $-b$   
 B.  $c$   
 C.  $m-b$   
 D.  $k-c$   
 E.  $c-k$



- 39 If  $x = 3$ ,  $y = 2$  satisfy the simultaneous equations  $\begin{cases} ax + by = 2 \\ bx - ay = 3 \end{cases}$ , find the values of  $a$  and  $b$ .

- A.  $a = 0$ ,  $b = 1$   
 B.  $a = 0$ ,  $b = -1$   
 C.  $a = \frac{5}{6}$ ,  $b = -\frac{1}{4}$   
 D.  $a = -\frac{1}{13}$ ,  $b = \frac{37}{39}$   
 E.  $a = -\frac{12}{13}$ ,  $b = \frac{5}{13}$

- 40 From the table, which of the following intervals must contain a root of  $f(x) - x = 0$ ?

| $x$ | $f(x)$ |
|-----|--------|
| -2  | 1.2    |
| -1  | 0.8    |
| 0   | 0.7    |
| 1   | 0.2    |
| 2   | -0.1   |
| 3   | 0.8    |

- A.  $-2 < x < -1$   
 B.  $-1 < x < 0$   
 C.  $0 < x < 1$   
 D.  $1 < x < 2$   
 E.  $2 < x < 3$

- 41 If the product of the first  $n$  terms of the sequence  $10, 10^2, 10^3, \dots, 10^n$  exceeds  $10^{55}$ , find the minimum value of  $n$ .

- A. 9  
 B. 10  
 C. 11  
 D. 12  
 E. 56

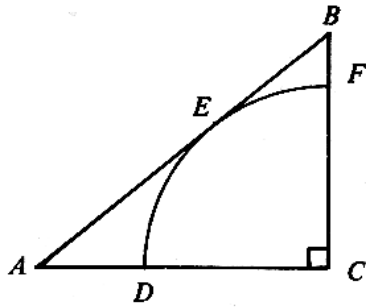
- 42 If  $a:b=2:3$ ,  $a:c=3:4$  and  $a:d=4:5$ ,  
then  $b:c:d=$

A.  $2:3:4$   
B.  $3:4:5$                       D.  $18:16:15$   
C.  $3:6:10$                       E.  $40:45:48$

- 43 Let  $x$  vary inversely as  $\sqrt{y}$ . If  $y$  is increased by 69%, then  $x$  will be

A. increased by 23.1%(3 sig. fig.)  
B. increased by 30%  
C. decreased by 23.1%(3 sig. fig.)  
D. decreased by 30%  
E. decreased by 76.9%(3 sig. Fig)

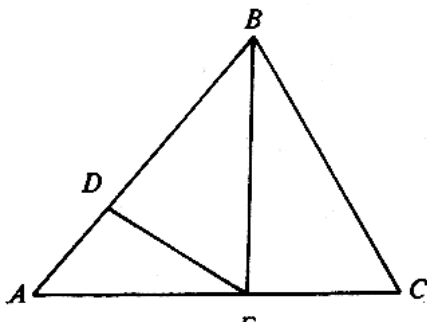
- 44 In the figure,  $CDEF$  is a sector of a circle which touches  $AB$  at  $E$ . If  $AB=25$  and  $BC=15$ , find the radius of the sector.



A. 9                                  D. 12  
B. 10                                E. 12.5  
C. 11.25

- 45 In the figure,  $AD:DB=1:2$ ,  $AE:EC=3:2$ .

Area of  $\triangle BDE$  : Area of  $\triangle ABC =$

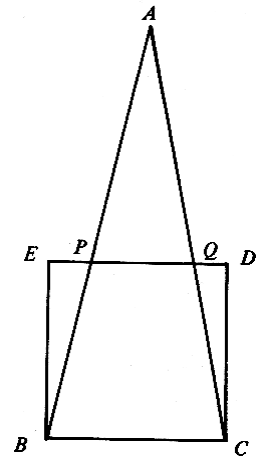


A.  $1:3$   
B.  $2:5$                           D.  $4:25$   
C.  $3:4$                           E.  $36:65$

- 46 In the figure, area of  $\triangle ABC$  : area of square

$BCDE = 2:1$ . Find  $PQ:BC$

A.  $1:2$   
B.  $1:3$   
C.  $1:4$   
D.  $2:3$   
E.  $3:4$



- 47 For  $0^\circ \leq x \leq 360^\circ$ , how many roots does the

equation  $\sin x(\cos x + 2) = 0$  have?

A. 0                                  D. 3  
B. 1                                E. 4  
C. 2

- 48 The largest value of  $(3\cos 2\theta - 1)^2 + 1$  is

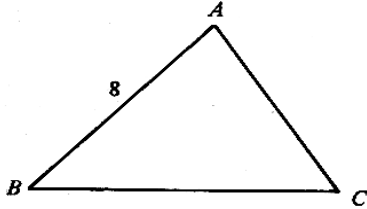
A. 2

- B. 5                      D. 26  
C. 17                     E. 50

49 In the figure,  $\sin A : \sin B : \sin C = 4 : 5 : 6$ .

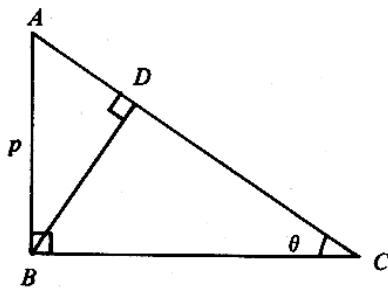
If  $AB=8$ , find  $AC$ .

- A.  $5\frac{1}{3}$   
B.  $6\frac{2}{3}$   
C.  $9\frac{3}{5}$   
D. 10  
E. 12

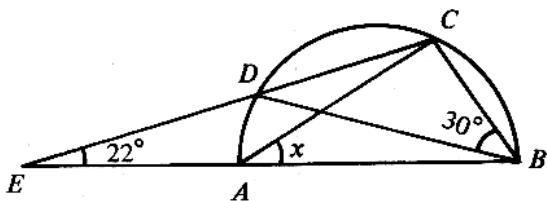


50 In the figure,  $AB=p$ ,  $\angle ACB=\theta$ . Find  $CD$ .

- A.  $p \sin \theta$   
B.  $p \cos \theta$   
C.  $\frac{p \sin \theta}{\cos^2 \theta}$   
D.  $\frac{p \sin^2 \theta}{\cos \theta}$   
E.  $\frac{p \cos^2 \theta}{\sin \theta}$



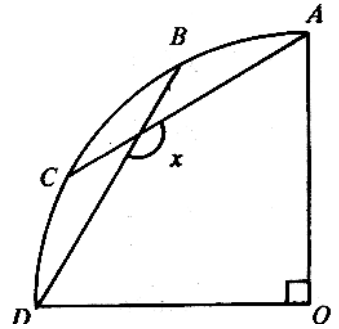
51 In the figure,  $ABCD$  is a semi-circle,  $CDE$  and  $BAE$  are straight lines. If  $\angle CBD=30^\circ$  and  $\angle DEA=22^\circ$ , find  $x$ .



52 In the figure,  $OABCD$  is a sector of a circle. If

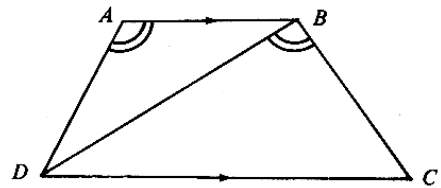
$\widehat{AB} = \widehat{BC} = \widehat{CD}$ , then  $x =$

- A.  $105^\circ$   
B.  $120^\circ$   
C.  $135^\circ$   
D.  $144^\circ$   
E.  $150^\circ$



53 In the figure,  $AB \parallel DC$  and  $\angle DAB = \angle DBC$ .

Which of the following is/are true ?



- I.  $\frac{AB}{BD} = \frac{BD}{DC}$   
II.  $\frac{AB}{BD} = \frac{AD}{BC}$   
III.  $\frac{AD}{BD} = \frac{BD}{CD}$

- A. I only  
B. II only  
C. III only  
D. I and II only  
E. II and III only